Solutions to Examination Questions

(10:00 am – 12:00 noon, Monday, 26th April 2021)

Communications II, Spring 2021

Instruction:

- This is an open-book test. Books, class notes, internet materials, and calculators are all allowed.
- It is not allowed to discuss with anybody else during the test. The student needs to take the test independently.
- Incorrect answers with justification and REASONING can receive partial credit. However, correct answer without justification or reasoning will receive no credit.
- Students need to answer all 4 problems.
- The following page has some formulas that can be used in your examination.

Prob I (*32 pt*)	
Prob II (*24 pt*)	
Prob III (*29pt*)	
Prob IV (*15 pt*)	
Total(*100pt*)	

Information for Students

Some Fourier Transform Pairs

	$x(t)=\int_{-\infty}^{\infty}X(f)e^{j2\pi ft}df$	$X(f)=\int_{-\infty}^{\infty}x(t)e^{-j2\pi ft}dt$
1	$x(t)=1$	$X(f)=\delta(f)$
2	$x(t)=\delta(t)$	$X(f)=1$
3	$x(t)=u(t)=\begin{cases}1, & t\geq 0\\ 0, & t<0\end{cases}$	$\frac{1}{2}\delta(f)+\frac{1}{j2\pi f}$
4	$x(t)=sgn(t)=\begin{cases}1, & t<0\\ -1, & t\geq 0\end{cases}$	$X(f)=\frac{1}{j\pi f}$
5	$x(t)=\frac{1}{\pi t}$	$X(f)=-jsgn(f)=\begin{cases}-j, & f<0\\ j, & f\geq 0\end{cases}$
6	$x(t)=rect\left(\frac{t}{T}\right)=\begin{cases}1, & \lvert t\rvert\leq T/2\\ 0, & \lvert t\rvert>T/2\end{cases}$	$X(f)=T\frac{\sin(\pi fT)}{\pi fT}=Tsinc(\pi fT)$
7	$x(t)=W\frac{\sin(\pi Wt)}{\pi Wt}=Wsinc(\pi Wt)$	$X(f)=rect\left(\frac{f}{W}\right)=\begin{cases}1, & \lvert f\rvert\leq W/2\\ 0, & \lvert f\rvert>W/2\end{cases}$
8	$x(t)\cos(2\pi f_o t)$	$\frac{1}{2}X(f-f_0)+\frac{1}{2}X(f+f_0)$
9	$x(t)\sin(2\pi f_o t)$	$\frac{1}{2j}X(f-f_0)-\frac{1}{2j}X(f+f_0)$

Some Trigonometry Formulas

$$\sin x\cos y=\frac{1}{2}[\sin(x+y)+\sin(x-y)] \qquad \sin x+\sin y=2\sin\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\cos x\sin y=\frac{1}{2}[\sin(x+y)-\sin(x-y)] \qquad \sin x-\sin y=2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$$

$$\sin x\sin y=\frac{1}{2}[\cos(x-y)-\cos(x+y)] \qquad \cos x+\cos y=2\cos\frac{x+y}{2}\cos\frac{x-y}{2}$$

$$\cos x\cos y=\frac{1}{2}[\cos(x+y)+\cos(x-y)] \qquad \cos x-\cos y=-2\sin\frac{x+y}{2}\sin\frac{x-y}{2}$$

Approximation of Q(x)

$$Q(x)=\frac{1}{\sqrt{2\pi}}\int_x^{+\infty}e^{-\frac{t^2}{2}}dt\approx\frac{1}{\sqrt{2\pi}x}e^{-\frac{x^2}{2}}$$

Problem I: Passband and Baseband Signal and Noise

a) Consider a passband noise, $N(t) = N_I(t)\cos(300\pi t) - N_Q(t)\sin(300\pi t)$, where $N_I(t)$ and $N_Q(t)$ are the corresponding in-phase and quadrature components and are independent and wide-sense stationary (WSS) Gaussian random processes with zero mean and the same power spectrum density. Denote the power spectrum density of $N_I(t)$ and $N_Q(t)$ as

$$S(f) = \begin{cases} 0.25, & |f| \le 2, \\ 0, & |f| > 2. \end{cases}$$

(**Hint:** $R(\tau) = \int_{-\infty}^{\infty} S(f)e^{j2\pi f\tau}df = sinc(4\pi\tau)$.)

1) [**3 points**] Find the variance of $N_I(t)$;

Since $E\{N_I(t)\} = 0$, $Var\{N_I(t)\} = E\{N_I^2(t)\} = \int_{-\infty}^{\infty} S(f)df = \int_{-2}^{2} 0.25df = 1$

2) [**3 points**] Prove $E\big(N(t)\big) = 0$;

$$\begin{aligned} E\{N(t)\} &= E\{N_I(t)\cos(300\pi t) - N_Q(t)\sin(300\pi t)\} \\ &= E\{N_I(t)\}\cos(300\pi t) - E\{N_Q(t)\}\sin(300\pi t) \\ &= 0 \times \cos(300\pi t) - 0 \times \sin(300\pi t) = 0 \end{aligned}$$

3) [**3 points**] Prove $E\left(N^2(t)\right) = 1$;

Since $N(t)$ has the same power as $N_I(t)$ and $N_Q(t)$, we have $E(N^2(t)) = E(N_I^2(t)) = 1$

4) [**3 points**] Find the probability density function of $N(0)$;

Since $N_I(t)$ and $N_Q(t)$ are Gaussian, so is $N(t)$. The probability density function of $N(0)$ is

$$f_{N(t)}(x;0) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}$$

5) [**5 points**] Prove that $N(t)$ is wide-sense stationary and find $E\{N(t)N(t+\tau)\}$;

Direct calculation yields that

$$E\{N(t)N(t+\tau\} = sinc(4\pi\tau)\cos(300\pi\tau)$$

Since $E\{N(t)\} = 0$ is independent of time and correlation depends only on the time difference, $N(t)$ is WSS.

6) [**3 points**] Find $S_{N(t)}(f)$, the power spectrum density of $N(t)$;

$$S_{N(t)}(f) = \int_{-\infty}^{\infty} sinc(4\pi\tau)\cos(300\pi\tau)e^{-j2\pi f\tau}d\tau = \frac{1}{2}\{S(f-150) + S(f+150)\}$$

7) [**2 points**] Find the joint probability density function of $N(t)$ and $N(t+1)$;

Since $E\{N(t)N(t+1\} = R(1)\cos(300\pi)=0$, we have

$$f_{N(t)}(x_1, x_2; t, t+1) = \frac{1}{\sqrt{2\pi}}e^{-\frac{x_1^2}{2}}\frac{1}{\sqrt{2\pi}}e^{-\frac{x_2^2}{2}} = \frac{1}{2\pi}e^{-\frac{x_1^2+x_2^2}{2}}$$

Problem I: Passband and Baseband Signal and Noise (Cont.)

b) Let $u(t) = \cos(210\pi t)$ be a passband signal.

1) [**5 points**] Find the corresponding in-phase and quadrature signals when the carrier frequency $f_c = 100$ Hz.

From down-conversion block diagram,

$$u(t)\, 2\cos(2\pi f_c t) = 2\cos(210\pi t)\cos(200\pi t) = \cos(10\pi t) + \cos(410\pi t)$$

After the low-pass filter, we get the in-phase term as

$$u_I(t) = \cos(10\pi t)$$

Similarly, we have

$$u(t)\,(-2\sin(2\pi f_c t)) = -2\cos(210\pi t)\sin(200\pi t) = \sin(10\pi t) - \sin(410\pi t)$$

After the low-pass filter, we can get the quadrature term as

$$u_Q(t) = \sin(10\pi t)$$

2) [**5 points**] Is it possible that the corresponding in-phase and quadrature signals are $u_I(t) = 1$, $u_Q(t) = 0$ for certain carrier frequency, f_c? Find the carrier frequency if it is possible, or explain why it is impossible.

Yes. It is possible.

If $f_c = 105$ Hz, then we can have $u_I(t) = 1$, $u_Q(t) = 0$ using similar procedures as part 1).

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Problem II: Analog Modulation

Consider an analog modulation system with carrier frequency $f_c = 500$ kHz. The message signal, $m(t)$, is wide-sense stationary (WSS) and with zero mean and the following power spectral density (PDF)

$$S_{m(t)}(f) = \begin{cases} \frac{1}{1000}\left(1 - \frac{|f|}{1000}\right), & |f| \le 1000 \text{ Hz}, \\ 0, & \text{otherwise.} \end{cases}$$

Therefore, the average power of the message signal will be $E\{|m(t)|^2\} = 1$. Assume $m_p = \max|m(t)| = 1.5$ and $A = 2$ and denote $T_c = \frac{1}{f_c}$.

a) For single-sideband (SSB) modulation, the modulated signal can be expressed as $s_{SSB}(t) = \frac{1}{2}Am(t)\cos(2\pi f_c t) - \frac{1}{2}A\,\hat{m}(t)\sin(2\pi f_c t)$, where $\hat{m}(t)$ is the Hilbert transform of $m(t)$. (**hint:** *Since* $\cos(2\pi f_c t)$ *and* $\sin(2\pi f_c t)$ *are orthogonal, you only need to find the average powers corresponding to each individual waveform and add them, that is,* $P_{SSB} = \frac{1}{T_c}\int_0^{T_c} E|s_{SSB}(t)|^2 dt = \frac{1}{T_c}\int_0^{T_c} E\left|\frac{1}{2}Am(t)\cos(2\pi f_c t)\right|^2 dt + \frac{1}{T_c}\int_0^{T_c} E\left|\frac{1}{2}A\,\hat{m}(t)\sin(2\pi f_c t)\right|^2 dt$)

1) [**4 points**] Find the power of the modulated signal, $P_{SSB} = \frac{1}{T_c}\int_0^{T_c} E|s_{SSB}(t)|^2 dt$.

$$P_{SSB} = \frac{1}{T_c}\int_0^{T_c} E\left|\frac{1}{2}Am(t)\cos(2\pi f_c t)\right|^2 dt + \frac{1}{T_c}\int_0^{T_c} E\left|\frac{1}{2}A\hat{m}(t)\sin(2\pi f_c t)\right|^2 dt$$

$$= \frac{1}{4}A^2\left(\frac{1}{T_c}\int_0^{T_c} E\{|m(t)|^2\}|\cos(2\pi f_c t)|^2 dt + \frac{1}{T_c}\int_0^{T_c} E\{|\hat{m}(t)|^2\}|\sin(2\pi f_c t)|^2 dt\right)$$

$$= \frac{1}{4}A^2\left(\frac{1}{T_c}\int_0^{T_c} |\cos(2\pi f_c t)|^2 dt + \frac{1}{T_c}\int_0^{T_c} |\sin(2\pi f_c t)|^2 dt\right) = \frac{1}{4}A^2 = \frac{1}{4}2^2 = 1$$

2) [**3 points**] Find the bandwidth of the modulated signal.

The bandwidth of the modulated signal for SSB is $W = 1000$ Hz.

b) For the standard amplitude modulation (AM), the modulated signal can be expressed as $s_{AM}(t) = (A + m(t))\cos(2\pi f_c t)$.

1) [**4 points**] Find the power of the modulated signal, $P_{AM} = \frac{1}{T_c}\int_0^{T_c} E|s_{AM}(t)|^2 dt$.

$$P_{AM} = \frac{1}{T_c}\int_0^{T_c} E|s_{AM}(t)|^2 dt = \frac{1}{T_c}\int_0^{T_c} E|A + m(t)|^2\cos^2(2\pi f_c t)\,dt$$

$$= E|A + m(t)|^2\frac{1}{T_c}\int_0^{T_c}\cos^2(2\pi f_c t)\,dt = E\left(A^2 + 2Am(t) + m^2(t)\right) \times \frac{1}{2} = (2^2 + 1) \times \frac{1}{2} = 2.5$$

2) [**3 points**] If the envelope detection is used, is there any distortion on the demodulated message signal? Why?

Since $A(= 2) \ge m_p(= 1.5)$, there is no distortion if using envelope detection.

3) [**3 points**] If coherent detection is used, justify whether there is any distortion.

No distortion if using coherent detection.

Problem II: Analog Modulation (Cont.)

c) For frequency modulation, the modulated signal can be expressed as $s_{FM}(t) = A\cos(2\pi f_c t + k_f \int_0^t m(\tau) d\tau)$, where the modulation index is $k_f = 3000$.

1) [**4 points**] Find the average power of the modulated signal.

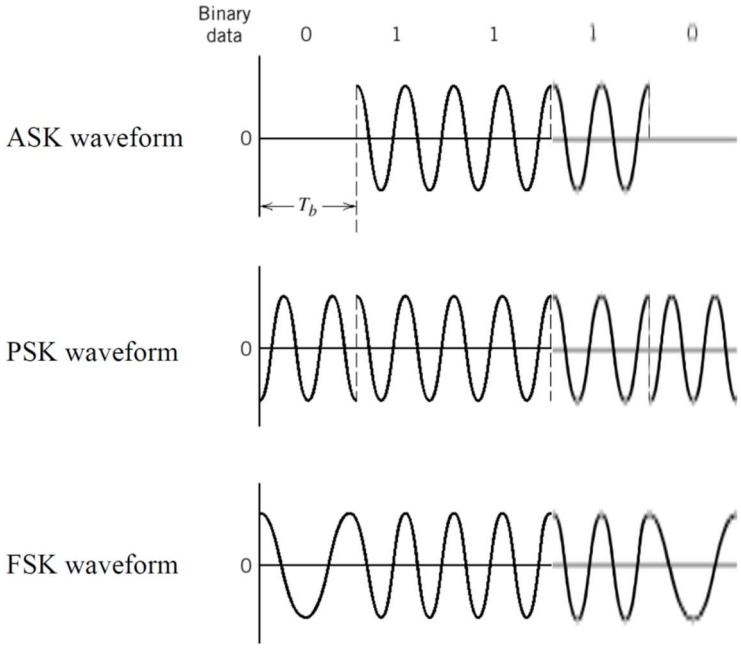
$$P_{FM} = \frac{1}{2}A^2 = \frac{1}{2}2^2 = 2$$

2) [**3 points**] Find the bandwidth of the modulated signal.

$B_T = 2(\Delta f + W) = 2(k_f m_p + W)$=2(3000× 1.5 + 1000)=11000 Hz

Problem III: Digital Modulation

a) The following are the waveforms for amplitude-shift keying (ASK), phase-shift keying (PSK), and frequency-shift keying (FSK), corresponding to the binary data, {0 1 1}. Please finish the waveforms if the binary data are {0 1 1 1 0}. [**3 point for each waveform**]



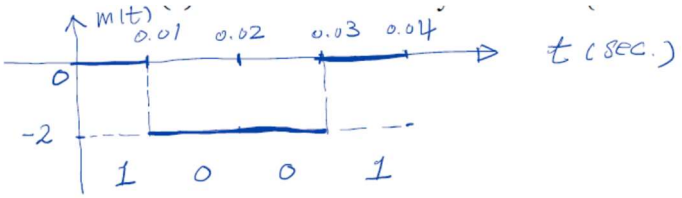
b) For ASK, the modulated signal can be expressed as

$$s(t) = \begin{cases} 0, & \text{if symbol 0 is transmitted,} \\ A\cos(2\pi f_c t), & \text{if symbol 1 is transmitted,} \end{cases}$$

where the amplitude, $A = 2$ volts, symbol duration is $T = 0.01\ sec$, and carrier frequency $f_c = 1000\ Hz$. We can express it into the form of standard amplitude modulation (AM) as $s(t) = (A + m(t))\cos(2\pi f_c t)$. Assume the standard deviation of noise, $\sigma = 0.4$ volts.

1) [**4 points**] Draw the waveform of $m(t)$ if the transmitted symbols are {1 0 0 1}.

If $b_n = 1$, then $m(t) = 0$; If $b_n = 0$, then $m(t) = -2$.



2) [**4 points**] Find the probability of error if coherent ASK detection is used.

$$\frac{A}{\sigma} = \frac{2}{0.4} = 5,\ P_{e,ASK} = Q\left(\frac{A}{2\sigma}\right) = Q(2.5) \approx \frac{1}{\sqrt{2\pi}\times 2.5} e^{-\frac{2.5^2}{2}} \approx 7.0 \times 10^{-3}$$

3) [**4 points**] Find the probability of error if noncoherent ASK detection is used.

$$P_{e,ASK} = \frac{1}{2} exp\left(-\frac{A^2}{8\sigma^2}\right) = \frac{1}{2} exp(-5^2/8) \approx 2.13 \times 10^{-2}$$

Problem III: Digital Modulation (Cont.)

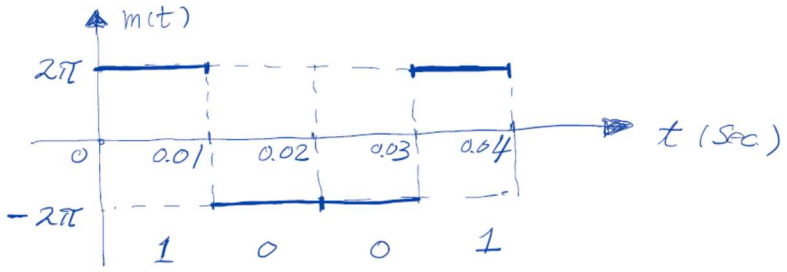
c) For FSK, the modulated signal can be expressed as

$$s(t) = \begin{cases} A\cos(2\pi f_0 t), & \text{if symbol 0 is transmitted}, \\ A\cos(2\pi f_1 t), & \text{if symbol 1 is transmitted}, \end{cases}$$

where the amplitude, $A = 2$ volts, symbol duration is $T = 0.01\ sec$, $f_0 = 900\ Hz$ and $f_1 = 1100\ Hz$. We can express FSK signal into the form of frequency modulation (FM) as $s(t) = A\cos(2\pi f_c t + k_f \int_0^t m(\tau)d\tau)$, where $k_f = 100$ and $f_c = 1000\ Hz$. Assume the standard deviation of noise, $\sigma = 0.4$ volts.

1) [**4 points**] Draw the waveform of $m(t)$ if the transmitted symbols are {1 0 0 1};

If $b_n = 1$, then $m(t) = 2\pi$; If $b_n = 0$, then $m(t) = -2\pi$.



2) [**4 points**] Find the probability of error if coherent FSK detection is used.

$\frac{A}{\sigma} = \frac{2}{0.4} = 5$

$$P_{e,FSK} = Q\left(\frac{A}{\sqrt{2}\sigma}\right) = Q\left(\frac{5}{\sqrt{2}}\right) \approx \frac{1}{\sqrt{2\pi} \times \left(\frac{5}{\sqrt{2}}\right)} e^{-\frac{\left(\frac{5}{\sqrt{2}}\right)^2}{2}} \approx 2.18 \times 10^{-4}$$

Problem IV: Basic Information Theory and Coding

a) Consider an information source with three symbols, a_1, a_2, and a_3, with $\Pr\{a_1\} = 0.6$, $\Pr\{a_2\} = 0.3$, and $\Pr\{a_3\} = 0.1$.

1) [**5 points**] Find the Huffman code and the corresponding average coding length;

There are multiple Huffman codes. Two of them are

$$a_1 \to 0, a_2 \to 10, a_3 \to 11; \text{ or } a_1 \to 1, a_2 \to 01, a_3 \to 00.$$

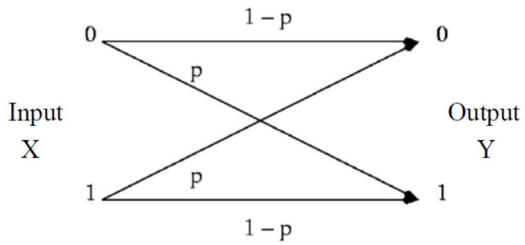
However, the average coding length should be same for all. It is

$$\bar{L} = 0.6 \times 1 + 0.3 \times 2 + 0.1 \times 2 = 1.4 \text{ bits}$$

2) [**2 points**] Does there exist a uniquely decodable code with the average length smaller than the entropy of the source? Find one if there exists, or explain if it does not exist.

No. According to the source coding theorem, there is no uniquely decodable code with the average length smaller than the entropy of the source.

b) For the following binary symmetric channel (BSC) with $p = 0.2$, the probabilities of the input symbols are $\Pr(X = 0) = \Pr(X = 1) = 0.5$. (**Hint:** *Denote* $h(q) = -q\log_2 q - (1-q)\log_2(1-q)$, *then* $h(0.1) = 0.47$, $h(0.2) = 0.72$, $h(0.3) = 0.88$, $h(0.4) = 0.97$, $h(0.5) = 1.0$)



1) [**4 points**] Find the mutual information between input and output symbols, $I(X,Y)$;

$I(X,Y) = H(Y) - H(Y|X) = 1 - h(p) = 1 - h(0.2) = 1 - 0.72 = 0.28$ bit

2) [**4 points**] Find the probability of error, that is, the probability that X and Y are different, $\Pr\{X \neq Y\}$.

$P_e = \Pr\{X \neq Y\}$

$= \Pr\{X \neq Y | X = 0\} \Pr\{X = 0\} + \Pr\{X \neq Y | X = 1\} \Pr\{X = 1\}$

$= \Pr\{Y = 1 | X = 0\} \Pr\{X = 0\} + \Pr\{Y = 0 | X = 1\} \Pr\{X = 1\}$

$= \mathrm{p} \times 0.5 + \mathrm{p} \times 0.5 = 0.2$